

CHAPTER 10

Energy changes in chemical reactions

When a piece of wood burns, we observe the release of energy mainly as heat and light. The energy is released to the surroundings and the burnt wood and ash will contain less energy than before it was burned. The energy that was once stored in the wood has been released to the environment. In a similar way, stored energy in the food that we eat can be released to provide us with energy that the organs in our bodies require to be able to function.

In this chapter you will learn about energy transformation processes like these; how seemingly 'invisible' energy from chemicals can be released to produce energy, and also how some chemical reactions absorb heat energy from the surroundings, actually increasing the amount of this stored 'chemical' energy.

By the end of this chapter, you will have a greater understanding of how energy changes occur as a result of chemical reactions, and be able to represent these changes in a range of ways, including chemical equations and energy diagrams.

Science understanding

- chemical reactions can be represented by chemical equations; balanced chemical equations indicate the relative numbers of particles (atoms, molecules or ions) that are involved in the reaction
- chemical reactions and phase changes involve enthalpy changes, commonly observable as changes in the temperature of the surroundings and/or the emission of light
- endothermic and exothermic reactions can be explained in terms of the Law of Conservation of Energy and the breaking of existing bonds and forming of new bonds; heat energy released or absorbed by the system to or from the surroundings, can be represented in thermochemical equations
- the mole is a precisely defined quantity of matter equal to Avogadro's number of particles

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10.1 Exothermic and endothermic reactions



FIGURE 10.1.1 The energy released by the combustion of wood in a fire is easily seen and felt.

Chemical reactions occur when particles (atoms, molecules or ions) collide and are rearranged to form new particles. Chemical reactions involve energy changes. As the reactant particles are rearranged, the chemical energy of the reactants is also changed.

In some chemical reactions, including the combustion of fuels, the rearrangement of atoms causes energy to be released to the surroundings (Figure 10.1.1). In other chemical reactions, energy is absorbed from the surroundings as the chemical reaction takes place. The energy change in some reactions is very small and can only be determined with specialised equipment.

In this section, you will learn about the energy changes that occur during chemical reactions. You will also learn how to classify chemical reactions based on their energy changes.

CHEMICAL ENERGY

There are many different forms of energy. For example, you are in contact with forms of heat (thermal energy), light, sound energy and electrical energy every day. You can probably see, hear or feel some of them as you are reading these pages.

All substances have a form of energy called **chemical energy**. It is stored in the chemical bonds between atoms and molecules. This energy results from:

- attractions between electrons and protons in atoms
- repulsions between nuclei
- repulsions between electrons
- movement of electrons
- vibrations and rotations around bonds.

However, unlike heat, light or sound energy, we cannot directly observe this energy, we only experience the effect of changes of this stored chemical energy. For example, when you eat a meal, the bonds between the food molecules have stored energy that you can access to provide energy for other chemical and physical activities that take place in your body (Figure 10.1.2).



FIGURE 10.1.2 When you eat food, you access the chemical energy stored in the food. This energy powers all of the chemical reactions and physical activities that take place in your body.

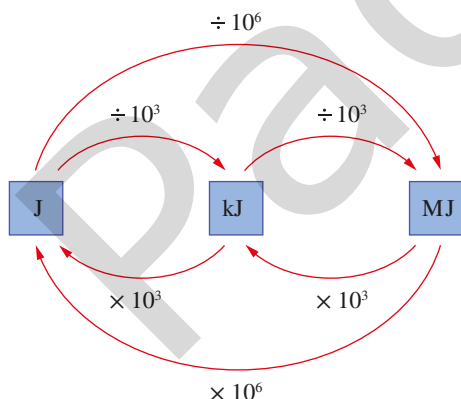


FIGURE 10.1.3 Converting between different energy units

The SI unit for energy is the joule, J. Other units also used for quantifying energy are kilojoules, kJ, and megajoules, MJ.

The relationship between joules, kilojoules and megajoules is:

$$1 \text{ MJ} = 1000 \text{ kJ} = 1\,000\,000 \text{ J}$$

Figure 10.1.3 shows how you can convert between different units of energy.

ENERGY CONSERVATION

The **law of conservation of energy** states energy cannot be created or destroyed. However, it can change forms. When energy is transformed from one form to another, the total amount of energy remains the same. So, if the chemical energy stored in a substance reduces, that energy must go somewhere. Likewise, a substance can't gain in chemical energy without absorbing that energy from another source.

Systems and surroundings

When we talk about energy changes in chemical reactions, we often refer to a system and its surroundings.

In chemistry, the **system** is usually the chemical reaction. When we say that energy is released or absorbed by a system, we are referring to energy changes that occur as bonds are broken and formed between the atoms of the elements involved in the reaction.

The **surroundings** are usually regarded as everything else. For example, the walls of a container in which a reaction takes place in the gas phase, or the water in a solution in which a reaction takes place, can be regarded as the surroundings for the reaction. Energy leaves the system (the reaction) and enters the surroundings, or leaves the surroundings and enters the system.

Energy changes during chemical reactions

The reactants in a chemical reaction have a certain amount of chemical energy stored in their bonds. The products that form as a result of the rearrangement of particles during the chemical reaction have different bonds and so have a different amount of chemical energy. Energy will be released or absorbed during the reaction depending on the relative energies of the bonds within the reactants and products.

In general, when particles are separated from each other, energy is required for this process to happen. The separated particles will therefore have more chemical energy than when they were together. This process can be seen in Figure 10.1.4. Conversely, if you consider particles coming together, the final arrangement with the particles joined, or bonded together, will have less energy than the separate particles. According to the law of conservation of energy, this 'lost' energy must go somewhere. In reality it is transferred to the surroundings. In Figure 10.1.4 this energy is represented by the wavy arrow, leaving the particles of the substance.

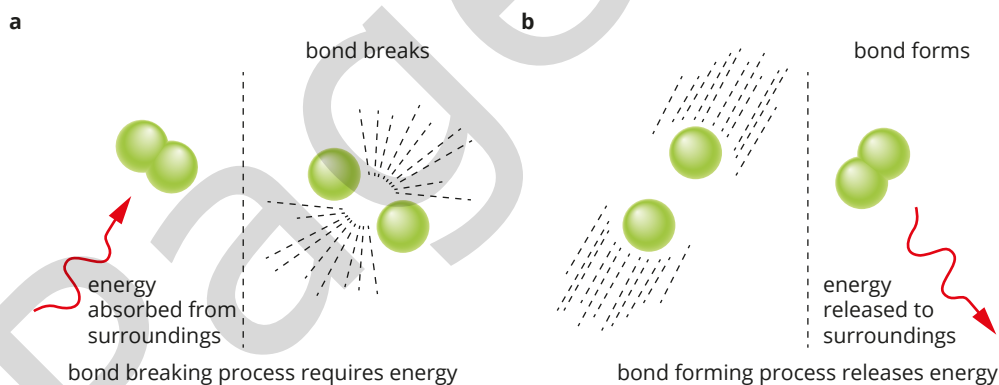


FIGURE 10.1.4 Energy changes when chemical bonds are broken and made

Often the energy released to or absorbed from the surroundings is in the form of heat (thermal energy). However, other forms of energy can also be released or absorbed including light, electricity and movement (kinetic energy).

CHEMISTRY IN ACTION

Glow-in-the-dark sticks

You might have seen glow-in-the-dark hoops, necklaces and bracelets similar to those shown in Figure 10.1.5 at festivals or concerts.

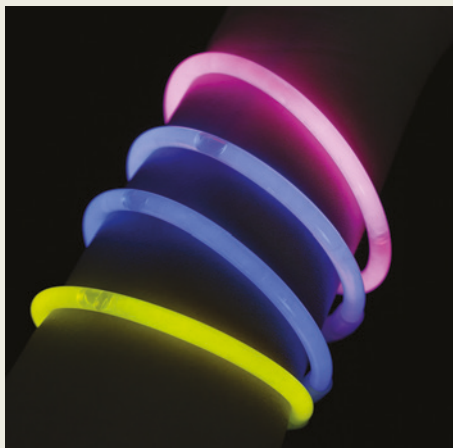


FIGURE 10.1.5 Glow-in-the-dark bracelets give off light that is the result of chemiluminescence.

Glow-in-the-dark bracelets contain chemicals held in separate containers. When these bracelets are bent, the containers break and the chemicals combine. Light is produced through a process called **chemiluminescence**. When the aqueous reactants (hydrogen peroxide in one container and diphenyl oxalate in another container) mix, energy is released from the reaction that occurs. This reaction is shown in Figure 10.1.6.

Instead of the energy from this reaction being released to the surroundings solely as heat, a carrier molecule transfers the energy to a **chemiluminescent** dye in the glow stick. The electrons in the dye are excited to higher energy levels. Light is emitted as these electrons return to their original lower energy levels. The light from the glow stick is the result of the emission spectrum (as discussed in Chapter 3) of the dye molecule.

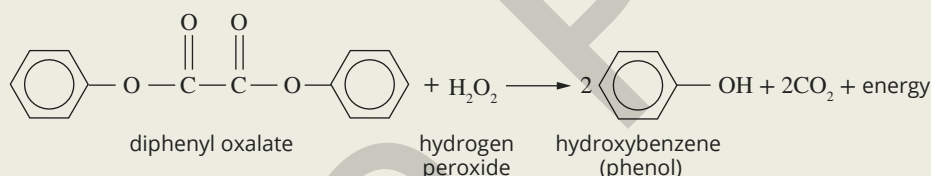


FIGURE 10.1.6 This reaction occurs in a glow stick.

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Glow-worms

Glow-worms (Figure 10.1.7) apply similar chemical principles to chemiluminescence for their glow-in-the-dark **bioluminescence**. Three chemicals within the worm combine. However, they require oxygen to produce light. When the worm breathes, oxygen acts as the oxidising agent in the chemical reaction between the three reactants producing the bioluminescence. Worms are able to control the amount of 'glow' by breathing in more or less oxygen.

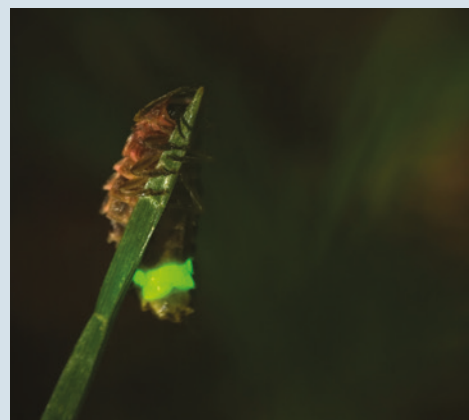


FIGURE 10.1.7 A female glow-worm. The luminescent abdominal organs are visible.

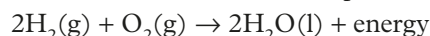
EXOTHERMIC AND ENDOTHERMIC SYSTEMS

Exothermic reactions

A chemical reaction in which energy is released is called an **exothermic** reaction. In this situation the total chemical energy of the products of the chemical reaction is less than the total chemical energy of the reactants. This 'lost' energy is released to the surroundings. Energy 'exits' the reaction system and the chemical reaction is called an exothermic reaction.

The released energy can be shown in a chemical equation by writing 'energy' on the product side of the arrow.

For example, the production of water from the reaction between hydrogen and oxygen gas is an exothermic reaction. This can be represented by the equation:



Another example of an exothermic reaction is the combustion of methane gas:



Heat is given off to the surroundings. All combustion reactions give off heat energy to the surroundings and are therefore exothermic reactions.

Endothermic reactions

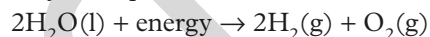
When the total chemical energy of the products of a chemical reaction is greater than the total chemical energy of the reactants, energy is absorbed from the surrounding environment. Energy 'enters' the reaction system and the chemical reaction is called an **endothermic** reaction.

If an endothermic reaction takes place in a container, the container may feel cold to the touch. This is because the reaction system is absorbing heat from the surroundings, leaving the environment cooler.

The idea that a process that absorbs energy gets cooler can be confusing, but to avoid this confusion you have to look at the system and the surroundings separately. The energy that is gained by the system is the 'unobservable' stored chemical energy that we cannot experience or directly measure. This energy is absorbed from the surroundings, which lose heat (i.e. get colder).

In a chemical equation of an endothermic reaction, the energy that is required can be written on the reactant side of the equation arrow.

For example, the **decomposition** of water by electrolysis (passing an electric current through the water) into hydrogen and oxygen is an endothermic process. This can be represented by the equation:



i If the total chemical energy of the products is *less* than the total energy of the reactants, energy will be released from the system into the surroundings. This is called an exothermic reaction.

If the total chemical energy of the products is *greater* than the total energy of the reactants, energy will be absorbed from the surroundings. This is called an endothermic reaction.

Changes of state

Changes of state (phase changes), such as a solid melting to form a liquid, are a type of physical change, rather than a chemical change. But these changes also involve energy being absorbed or released.

For example, the melting of ice into water requires the absorption of energy, making it an endothermic process. The boiling of water to produce water vapour is also endothermic.

Conversely, condensing a gas to a liquid and freezing a liquid to form a solid both release heat to the surroundings and so are exothermic processes. These processes are summarised in Figure 10.1.8.

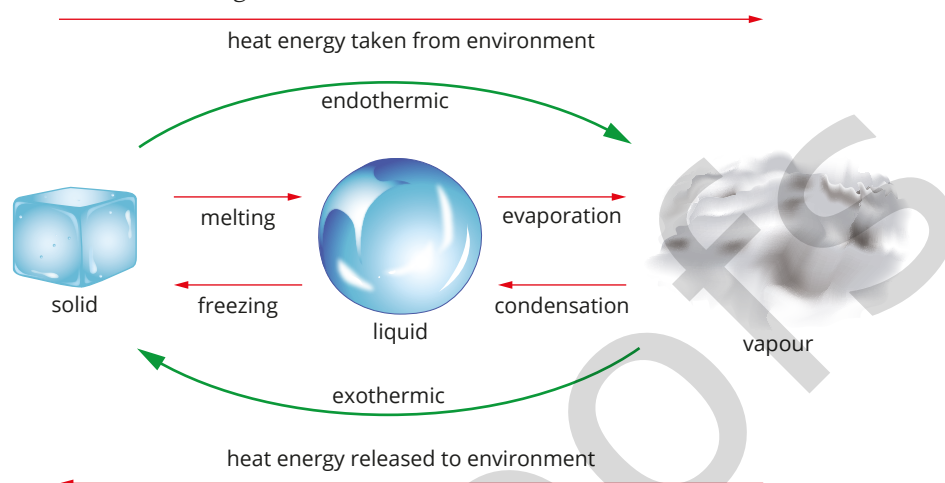
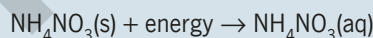


FIGURE 10.1.8 Energy changes associated with phase changes

CHEMFILE

Instant cold packs

Instant cold packs are often carried in first aid kits at sporting events (Figure 10.1.9). One type of cold pack contains a sealed bag of water surrounded by solid ammonium nitrate. When the cold pack is squeezed, the water bag is broken and ammonium nitrate dissolves in the water. The process is endothermic, absorbing heat from the surroundings and quickly lowering the pack's temperature:



Notice that bonds (in this case ionic bonds) are being broken, hence energy is required to be absorbed from the surroundings. Writing this process as an ionic equation emphasises this point:

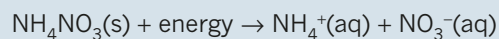


FIGURE 10.1.9 An endothermic reaction produces an instant cold pack.

10.1 Review

SUMMARY

- Energy is measured in J, kJ or MJ:
 $1 \text{ MJ} = 10^3 \text{ kJ} = 10^6 \text{ J}$
- Chemical energy is stored in the bonds between atoms.
- Energy is conserved during a chemical reaction; energy cannot be created or destroyed.
- Chemical reactions and changes of state involve energy changes.
- A chemical reaction that releases energy to the surroundings is called an exothermic reaction.
- A chemical reaction in which energy is absorbed from the surroundings is called an endothermic reaction.
- All combustion reactions are exothermic reactions.

KEY QUESTIONS

- Which of the following statements about combustion reactions is correct?
 - Combustion reactions are usually exothermic, meaning energy is absorbed during the reaction.
 - Combustion reactions are always exothermic, meaning energy is absorbed during the reaction.
 - Combustion reactions are usually endothermic, meaning energy is absorbed by the system.
 - Combustion reactions are always exothermic, meaning energy is released during the reaction.
- Convert the following energy values to kJ.
 - 0.180 MJ
 - $1.5 \times 10^6 \text{ J}$
 - 10.0 J
 - $2.0 \times 10^{-3} \text{ J}$
- Explain the difference between the terms 'system' and 'surroundings' in relation to a chemical reaction.
- Explain the term 'endothermic' in relation to the total amount of chemical energy of the reactants and products.

10.2 Thermochemical equations, energy profile diagrams and enthalpy

As you saw in the previous section, the word ‘energy’ can be included in a combustion equation to show that energy is released. However, it is generally more useful for a chemist to be precise about the magnitude (size) of the energy change that takes place.

Thermochemical equations achieve this by including a sign and numerical value for the energy change that occurs in the reaction.

It is also useful for chemists to be able to show the energy changes that occur as a reaction proceeds. All reactions absorb some energy before they can proceed, even if energy is released overall. An **energy profile diagram** represents the energy changes that occur during the course of a reaction.

In this section, you will learn how to write and interpret thermochemical equations and draw energy profile diagrams. These representations are important for the rates of reactions topic later in Year 11, and when considering chemical equilibrium in Year 12.

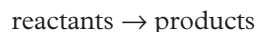
However, before we consider this, we need to give a name to this stored chemical energy that we have been discussing in section 10.1.

ENTHALPY

The stored chemical energy of a substance is called its **enthalpy**. It is given the symbol H . All chemicals have a certain amount of stored chemical energy, or enthalpy. However, it is the change in enthalpy that occurs in a chemical reaction that is most important to consider.

The exchange of heat energy between the system and its surroundings is referred to as the **enthalpy change**, or **heat of reaction**, and is given the symbol ΔH .

For a chemical reaction:



the enthalpy change (ΔH) is calculated by:

$$\Delta H = H_{\text{products}} - H_{\text{reactants}}$$

Enthalpy change in exothermic reactions

When the enthalpy of the products is less than the enthalpy of the reactants, energy is released from the system into the surroundings, so the reaction is exothermic. The system has lost energy, so ΔH has a negative value.

Therefore, for combustion reactions (which are exothermic reactions), $\Delta H < 0$ (Figure 10.2.1).

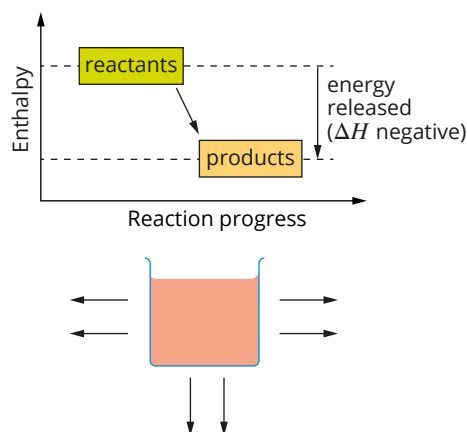


FIGURE 10.2.1 In the combustion of a fuel, the enthalpy of the reactants is greater than the enthalpy of the products, so energy is released to the surroundings during the reaction.

i Enthalpy change is a measure of the amount of energy absorbed or released during chemical reactions. It is given the symbol ΔH and is determined by subtracting the enthalpy of the reactants from the enthalpy of the products.

Explosives—extreme exothermic reactions

Humans have been using chemicals to make explosions since 919 BCE. Chinese people first mixed saltpetre (potassium nitrate), sulfur and charcoal with explosive results. They quickly realised that there were many uses for this mixture, which later became known as gunpowder. It was put to military use and eventually led to the development of bombs, cannons and guns.

Today, explosives are an essential tool for mining and other engineering works, such as road construction, tunnelling, building and demolition (Figure 10.2.2).

Explosives transform chemical energy into large quantities of thermal energy very quickly in highly exothermic reactions with a large drop in the enthalpy of the chemicals involved. Although thermal energy is also released when fuels such as petrol and natural gas burn, the rate of combustion in these reactions is limited by the availability of oxygen gas to the fuel. In contrast, the compounds making up an explosive contain sufficient oxygen for a complete (or almost complete) reaction to occur very quickly.



FIGURE 10.2.2 An old bridge is demolished with the help of explosives.

Enthalpy change in endothermic reactions

When the total enthalpy of the products is greater than the total energy of the reactants, energy must be absorbed from the surroundings, so the reaction is endothermic. The system has gained energy, so ΔH has a positive value, i.e. $\Delta H > 0$ (Figure 10.2.3).

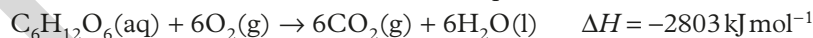
Thermochemical equations

The enthalpy change can be shown by writing the ΔH value to the right of the chemical equation. Such an equation is called a thermochemical equation. The ΔH value in a thermochemical equation usually has the units kJ mol^{-1} (kilojoule per mole). This means that the amount of energy (in kJ) signified by the ΔH value corresponds to the mole amounts specified by the coefficients in the equation.

For example, respiration in most living things can be considered as a type of combustion reaction of glucose. Therefore, respiration is an exothermic reaction. Previously you have seen such situations written with the energy included in the equation:



This can be written as the thermochemical equation shown below:



This thermochemical equation tells you that when 1 mole of glucose reacts with 6 moles of oxygen to produce 6 moles of carbon dioxide and 6 moles of water, 2803 kJ of energy is released to the surroundings.

The equation may also be written to include the enthalpy in the equation.



The ΔH is *still* equal to $-2803 \text{ kJ mol}^{-1}$.

Note that the energy is written on the right-hand side of the equation as a ‘product’ in this exothermic reaction.

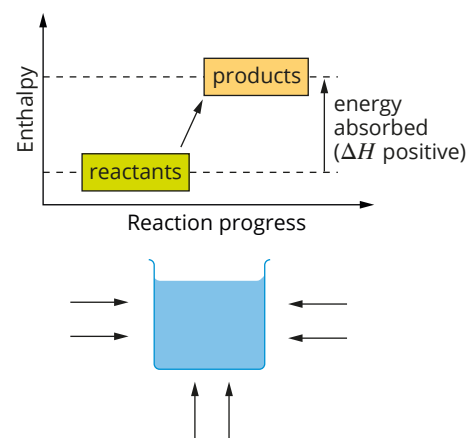
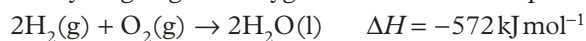


FIGURE 10.2.3 For an endothermic reaction, the enthalpy of the reactants is less than the enthalpy of the products so energy is absorbed during the reaction.

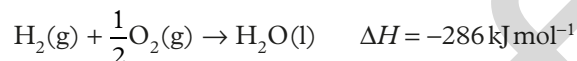
Enthalpy changes per mole of reactant

The combustion of hydrogen gas in oxygen is shown in the equation:



However, in this reaction there are 2 moles of hydrogen, and if we want to compare enthalpy changes, we have to ensure that we are comparing equal quantities of the substance, in this case 1 mole of the chemical being combusted, i.e. we must use kJ per 1 mole (kJ mol^{-1}).

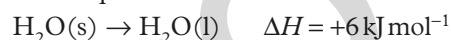
In this case we divide the coefficients in the equation by 2 so that there is only 1 mol of H_2 and the equation becomes:



Therefore the heat of combustion of hydrogen is given as -286 kJ mol^{-1} .

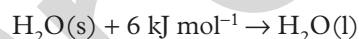
Physical changes

Enthalpy changes also occur during physical changes, so thermochemical equations can be written for physical changes. Melting ice is an example of a physical change. It is an endothermic process, because heat must be applied to solid ice in order to break the bonds between the water molecules when it is converted into liquid water. The thermochemical equation for this reaction is:



The ΔH value is positive because this is an endothermic reaction.

The equation may also be written to include the enthalpy in the equation.



The ΔH is still equal to $+6 \text{ kJ mol}^{-1}$.

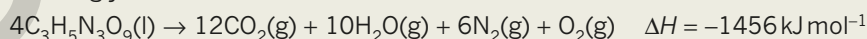
Note that it is very important to always include state symbols in all thermochemical equations. Physical changes involve an enthalpy change, so the state of the species in a chemical reaction will affect the enthalpy change of the reaction.

CHEMISTRY IN ACTION

The chemistry behind an explosion

When chemical explosives, such as ammonium nitrate, trinitrotoluene (TNT) and nitroglycerine decompose, they release large amounts of energy very quickly.

This is the thermochemical equation for the decomposition of nitroglycerine:



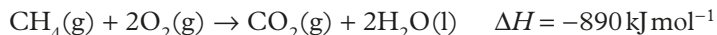
The negative ΔH value indicates that this is an exothermic reaction.

Notice that 29 moles of gas (the total number of moles of all products) are produced from 4 moles of nitroglycerine. At atmospheric pressure, the reactant products would expand to fill a volume more than 10 000 times larger than the volume of the nitroglycerine! During a blast, this gas is usually produced within a small hole into which the explosive has been placed, creating huge pressures that shatter the surrounding rock or structure.

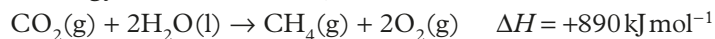
Effect on ΔH of reversing a chemical reaction

Reversing a chemical equation changes the sign but not the magnitude of ΔH .

For example, methane (CH_4) reacts with oxygen gas to produce carbon dioxide gas and water in an exothermic reaction:



If this reaction is reversed, the magnitude of ΔH remains the same because the enthalpies of the individual chemicals have not changed, but the sign changes to indicate that energy must be absorbed, i.e. it is an endothermic reaction:



This demonstrates the law of conservation of energy and is true for all reversible reactions.

Worked example 10.2.1

INTERPRETING A THERMOCHEMICAL REACTION

The ΔH of the following reaction is $-2220 \text{ kJ mol}^{-1}$. $\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g})$ Rewrite the reaction, including the energy term.	
Thinking	Working
The reaction is exothermic so show energy is a product in the reaction.	Energy term goes on right-hand side.
$\Delta H = -2220 \text{ kJ mol}^{-1}$ so add this value.	$\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g}) + 2220 \text{ kJ mol}^{-1}$

Worked example: Try yourself 10.2.1

INTERPRETING A THERMOCHEMICAL REACTION

The ΔH of the photosynthesis reaction shown below is 2803 kJ mol^{-1} . $6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6(\text{aq}) + 6\text{O}_2(\text{g})$ Rewrite the reaction, including the energy term.	
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Worked example 10.2.2

IDENTIFYING THE ENTHALPY CHANGE IN A THERMOCHEMICAL REACTION

Rewrite the reaction below, showing the enthalpy change separately. $\text{N}_2(\text{g}) + 2\text{O}_2(\text{g}) + 33 \text{ kJ mol}^{-1} \rightarrow 2\text{NO}_2(\text{g})$	
Thinking	Working
The reaction is endothermic—energy is gained from the surroundings.	Enthalpy change is positive.
Energy absorbed = 33 kJ mol^{-1}	$\text{N}_2(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g}) \quad \Delta H = +33 \text{ kJ mol}^{-1}$

Worked example: Try yourself 10.2.2

IDENTIFYING THE ENTHALPY CHANGE IN A THERMOCHEMICAL REACTION

Rewrite the reaction, showing the enthalpy change separately. $4\text{C}_3\text{H}_5\text{N}_3\text{O}_9(\text{l}) \rightarrow 12\text{CO}_2(\text{g}) + 10\text{H}_2\text{O}(\text{g}) + 6\text{N}_2(\text{g}) + \text{O}_2(\text{g}) + 1456 \text{ kJ mol}^{-1}$	
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i A reaction cannot proceed unless the bonds in the reactants are broken.

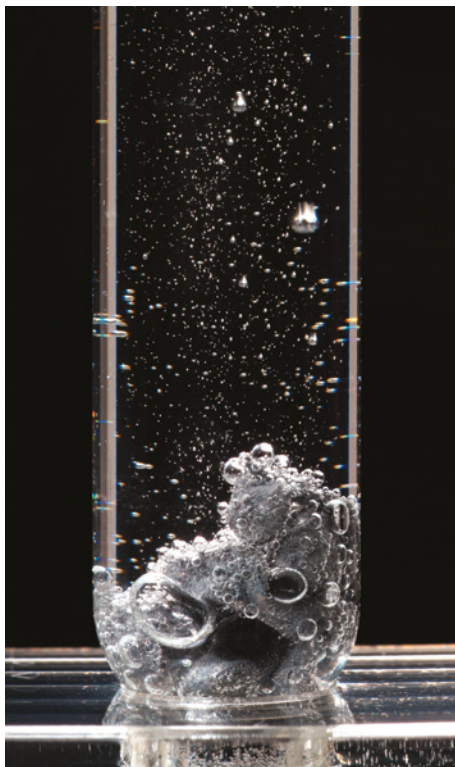


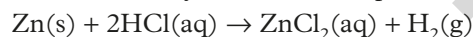
FIGURE 10.2.4 When zinc comes into contact with hydrochloric acid, it reacts almost immediately. The reactants have sufficient energy to 'overcome' the activation energy barrier.

Activation energy

The energy required to break the bonds of reactants so that a reaction can proceed is called the **activation energy**. The activation energy is an energy barrier that must be overcome before a reaction can get started. (The concept of activation energy is discussed in more detail in Chapter 18.)

An activation energy barrier exists for both exothermic and endothermic reactions. If the activation energy for a reaction is very low, the chemical reaction can be initiated as soon as the reactants come into contact because the reactants already have sufficient energy for a reaction to take place. An example of this can be seen in the reaction between zinc and hydrochloric acid in Figure 10.2.4.

The reaction between zinc and hydrochloric acid produces hydrogen gas:



As you can see in Figure 10.2.4, bubbles of hydrogen gas are vigorously produced as soon as zinc is added to the acid.

ENERGY PROFILE DIAGRAMS

The energy changes that occur during the course of a chemical reaction can be shown on an energy profile diagram.

The energy profile diagram for an exothermic combustion reaction like the one shown in Figure 10.2.5 indicates that the enthalpy of the products is always less than the enthalpy of the reactants. Overall, energy is released and so the ΔH value is negative. The energy profile also shows that, even in exothermic reactions, the activation energy must first be absorbed to start the reaction.

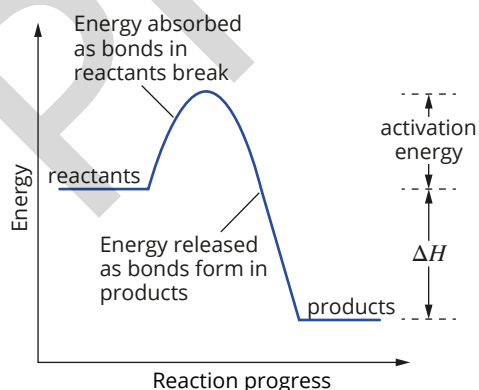


FIGURE 10.2.5 The characteristic shape of an energy profile diagram for an exothermic reaction

The energy profile diagram for an endothermic reaction (Figure 10.2.6) shows that the enthalpy of the products is greater than the enthalpy of the reactants. Overall, energy is absorbed and so the ΔH value is positive. The energy profile also shows the absorption of the activation energy before the release of energy as bonds form in the products.

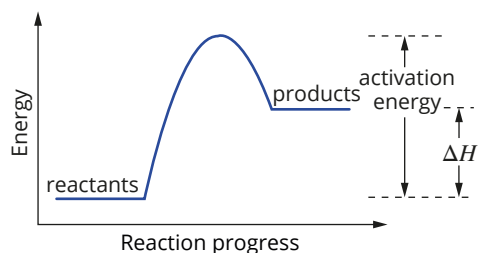


FIGURE 10.2.6 The characteristic shape of an energy profile diagram for an endothermic reaction

10.2 Review

SUMMARY

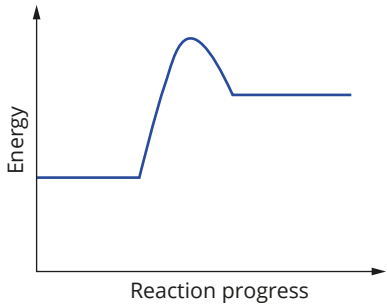
- Thermochemical equations include a ΔH value for a chemical reaction. The unit of ΔH is kJ mol^{-1} .
- The value of ΔH indicates the magnitude of the energy change and whether the energy is absorbed (a positive value) or released (a negative value).
- Reversing an equation causes the sign, but not the magnitude, of ΔH to change, as the reaction changes from exothermic to endothermic, or vice versa.
- States of matter must be included in thermochemical equations because changes of state involve enthalpy changes.
- Activation energy is the energy that must be absorbed to break the bonds in the reactants so that a chemical reaction can proceed. Both endothermic and exothermic reactions require activation energy.
- Energy profile diagrams show energy changes over the course of a reaction. They show the relative enthalpies of reactants and products and the activation energy.

KEY QUESTIONS

- Explain what a negative ΔH value indicates about a chemical reaction, in terms of the relative enthalpies of the reactants and products.
- When 1 mole of methane gas undergoes combustion in oxygen to produce carbon dioxide and water, 890 kJ of energy is released. Write a balanced thermochemical equation for this reaction.
- The combustion of octane to form carbon dioxide and liquid water can be written as:
$$\text{C}_8\text{H}_{18}(\text{g}) + 12\frac{1}{2}\text{O}_2 \rightarrow 8\text{CO}_2(\text{g}) + 9\text{H}_2\text{O}(\text{l})$$
$$\Delta H = -5450 \text{ kJ mol}^{-1}$$

The combustion of octane to form carbon dioxide and steam can be written as:

$$\text{C}_8\text{H}_{18}(\text{g}) + 12\frac{1}{2}\text{O}_2(\text{g}) \rightarrow 8\text{CO}_2(\text{g}) + 9\text{H}_2\text{O}(\text{g})$$

How would the energy released by the combustion of 1 mole of octane to form steam compare with the energy released by 1 mole of octane to form liquid water?
- The energy profile diagram below shows the energy changes during the reaction of hydrogen and iodine to form hydrogen iodide:
$$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$$

 - Is the reaction exothermic or endothermic?
 - Describe the relative enthalpies of the reactants and products.
 - Comment on the size of the activation energy compared with ΔH .
- The reaction for photosynthesis is the opposite of the reaction for respiration shown on page xxx. Write a thermochemical equation for photosynthesis.

Chapter review

KEY TERMS

activation energy	energy profile diagram	law of conservation of energy
chemical energy	enthalpy	surroundings
chemiluminescence	enthalpy change	system
chemiluminescent	exothermic	thermochemical equation
decomposition	heat of reaction	
endothermic		

Exothermic and endothermic reactions

- Convert the following units of energy to the unit given (to 3 significant figures).
 - 2205 J to kJ
 - 0.152 kJ to J
 - 1890000 J to MJ
 - 0.0125 MJ to kJ
- Decide whether the following processes are exothermic or endothermic. Give reasons for your answers.
 - burning of wood
 - melting of ice
 - recharging of a car battery
 - decomposition of plants in a compost heap
- Explain why chemiluminescence is an example of an exothermic reaction.

Thermochemical equations, energy profile diagrams and enthalpy

- Which one of the following statements is correct about the energy profile diagrams of both endothermic and exothermic reactions?
 - There is always less energy absorbed than released.
 - The enthalpy of the products is always less than the enthalpy of the reactants.
 - Some energy is always absorbed to break bonds in the reactants.
 - The ΔH value is the difference between the enthalpy of the reactants and the highest energy point reached on the energy profile.
- Identify whether each of the following statements related to activation energy is true or false.
 - Activation energy is the energy required to break bonds in the reactants.
 - Reactions that start immediately do not have an activation energy.
 - Reactions that release energy overall do not need to absorb activation energy.
 - The match used to light a fire is providing activation energy.
- Explain why reversing a chemical reaction reverses the sign of ΔH .

Connecting the main ideas

- The combustion of butane gas in portable stoves can be represented by the thermochemical equation:
$$2\text{C}_4\text{H}_{10}(\text{g}) + 13\text{O}_2(\text{g}) \rightarrow 8\text{CO}_2(\text{g}) + 10\text{H}_2\text{O}(\text{l})$$
$$\Delta H = -5772 \text{ kJ mol}^{-1}$$
 - How does the overall energy of the bonds in the reactants compare with the overall energy of the bonds in the products?
 - Draw an energy profile diagram for the reaction, labelling ΔH and activation energy.
- The enthalpy change for the combustion of 1 mole of hydrogen is -286 kJ mol^{-1} . Write a thermochemical equation for the complete combustion of hydrogen.
- In a steelworks, carbon monoxide present in the exhaust gases of the blast furnace can be used as a fuel elsewhere in the plant. It reacts according to the equation:
$$\text{CO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) \quad \Delta H = -283 \text{ kJ mol}^{-1}$$
 - Which has the greater total enthalpy: 1 mol of $\text{CO}(\text{g})$ and 0.5 mol of $\text{O}_2(\text{g})$, or 1 mol of $\text{CO}_2(\text{g})$?
 - Write the value of ΔH for the following equations.
 - $2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g})$
 - $2\text{CO}_2(\text{g}) \rightarrow 2\text{CO}(\text{g}) + \text{O}_2(\text{g})$